

Question 1

a)

First we see that $\psi(0) = 0$. Now let $z_1 = a + ib$ and $z_2 = c + id$. Then

$$z_1 + z_2 = (a + c) + i(b + d),$$

so

$$\psi(z_1 + z_2) = (a + c) + (b + d) \pmod{2}.$$

On the other hand,

$$\psi(z_1) + \psi(z_2) = (a + b) + (c + d) \pmod{2}.$$

By associativity and commutativity of addition modulo 2, these expressions are equal, hence ψ preserves addition. It follows from the two facts above, that ψ is a group homomorphism.

We also have $\psi(1) = 1$, so the multiplicative identity is preserved. For multiplication, we have

$$z_1 z_2 = (ac - bd) + i(ad + bc).$$

Applying ψ gives

$$\psi(z_1 z_2) = (ac - bd) + (ad + bc) \pmod{2} = ac + bd + ad + bc \pmod{2}.$$

Meanwhile,

$$\psi(z_1)\psi(z_2) = (a + b)(c + d) \pmod{2} = ac + ad + bc + bd \pmod{2}.$$

By commutativity of addition modulo 2, these expressions are equal. Thus ψ preserves multiplication.

Therefore ψ is a ring homomorphism.

b)

The kernel of ψ is given by

$$\ker(\psi) = \{z = a + ib \in \mathbb{Z}[i] \mid a + b = 0 \pmod{2}\}.$$

That is, every gaussian integer where the sum of the real and imaginary parts is even. See Figure 1.

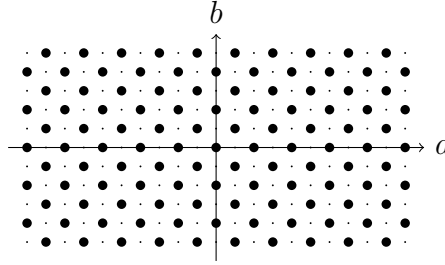


Figure 1: $\mathbb{Z}[i]$. Larger dots represent elements in the kernel of ψ .

c)

To use the ring isomorphism theorem, we must first show that $\langle 1+i \rangle$ is equal to the kernel of ψ .

First we show that $\langle 1+i \rangle$ is contained in the kernel of ψ . Take any $z \in \langle 1+i \rangle$. By definition of ideals, there exists a $k \in \mathbb{Z}[i]$ such that

$$z = (1+i) \cdot k.$$

Let $k = m + in$, where $m, n \in \mathbb{Z}$. Then

$$z = (1+i)(m+in) = (m-n) + i(m+n).$$

Applying ψ gives

$$\psi(z) = (m-n) + (m+n) \pmod{2} = 2m \pmod{2} = 0.$$

Therefore $z \in \ker(\psi)$, and since z was arbitrary, we have shown that $\langle 1+i \rangle \subseteq \ker(\psi)$.

Now we show that $\ker(\psi)$ is contained in $\langle 1+i \rangle$. Take any $z \in \ker(\psi)$. Let $z = a + ib$, where $a, b \in \mathbb{Z}$. Then, by definition of the kernel, we have

$$\psi(z) = a + b \pmod{2} = 0,$$

so $a + b$ is even. This means that either both a and b are even, or both are odd.

Now divide $a + ib$ by $1 + i$:

$$\frac{a+ib}{1+i} = \frac{(a+ib)(1-i)}{(1+i)(1-i)} = \frac{(a+b) + i(b-a)}{2} = \frac{a+b}{2} + i\frac{b-a}{2}.$$

Since a and b are both even or both odd, both $\frac{a+b}{2}$ and $\frac{b-a}{2}$ are integers. Therefore, $\frac{a+ib}{1+i} \in \mathbb{Z}[i]$, so $z = a + ib$ is a multiple of $1 + i$ in $\mathbb{Z}[i]$. Thus, $z \in \langle 1+i \rangle$, and since z was arbitrary, we have shown that $\ker(\psi) \subseteq \langle 1+i \rangle$.

Combining the two inclusions, we have shown that $\ker(\psi) = \langle 1 + i \rangle$. We can now use the ring isomorphism theorem, since we have established that ψ is a ring homomorphism from $\mathbb{Z}[i]$ to $\mathbb{Z}_2$ with kernel $\langle 1 + i \rangle$ and image $\mathbb{Z}_2$, to conclude that

$$\mathbb{Z}[i]/\langle 1 + i \rangle \simeq \mathbb{Z}_2.$$

d)

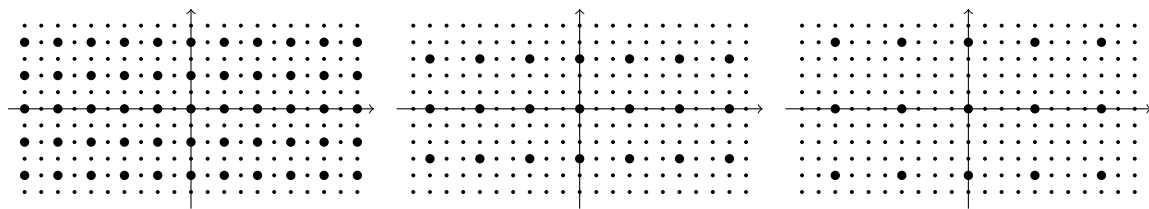


Figure 2: Ideals $\langle 2 \rangle$, $\langle 3 \rangle$, and $\langle 4 \rangle$ in $\mathbb{Z}[i]$. Highlighted points represent elements in the respective ideal.

Intuitively, each ideal $\langle n \rangle$ can be thought of as a grid of Gaussian integers where every n 'th integer in both the real and imaginary direction are included, corresponding to scaling the plane by a factor of n (see Figure 2). Working modulo the ideal then corresponds to working modulo n in both directions, giving n^2 distinct cosets.

More formally, any element $a + ib + \langle n \rangle$ in the quotient ring $\mathbb{Z}[i]/\langle n \rangle$ can be represented by the coset representative $r + is + \langle n \rangle$, where r and s are the remainders when a and b are divided by n , respectively. Since there are n possible remainders (from 0 to $n - 1$) for both r and s , there are a total of n^2 distinct cosets in the quotient ring.

Question 2

a)

We use division with remainder in $\mathbb{Z}_3[x]$ (steps not shown) to find that:

$$x^6 + 2x^4 + x^3 + x^2 + x = (x^2 + 2x)(x^4 + x^3 + x + 2)$$

Since the representative is a multiple of the generator, it is in the ideal. Therefore, the standard form of the coset is $0 + I$.

b)

We use the extended Euclidean algorithm in $\mathbb{Z}_5[x]$ (steps not shown) to find that:

$$3 \cdot (x^4 + x^3 + x + 2) + (2x^3 + 2x^2 + 2) \cdot x = 1$$

Since we are working modulo the ideal, this is equivalent to:

$$(2x^3 + 2x^2 + 2) \cdot x = 1$$

And indeed, multiplying out (modulo the ideal) gives:

$$2x^4 + 2x^3 + 2x = 2(x^4 + x^3 + x + 2) + 1 = 1$$

So the multiplicative inverse of $x + I$ in the quotient ring is $2x^3 + 2x^2 + 2 + I$.

c)

From the factorization given in the assignment, we have two zero divisors:

$$\begin{aligned} p(x) &= x + 4 \\ q(x) &= x^3 + 2x^2 + 2x + 3 \end{aligned}$$

They are zero divisors since they are their own GCD with the generator polynomial, and they each have degree less than 4 and greater than 0 (also, when multiplied together, they give the generator polynomial, which is zero in the quotient ring). If we now take $x \cdot p(x)$, this is also a zero divisor, since the GCD of $xp(x)$ and the generator polynomial is still $p(x)$.

d)

No, since ideals have to be closed under addition. Both $-x^2$ and $x^2 + 1$ are in the ideal, but their sum 1 is not.